Ioannis D. Gialamas

Curriculum Vitae

Personal Details

Date of Birth	September 29, 1993
Place of birth	Athens
Nationality	Greek
Compulsory Military Ser.	20/9/21 – 20/6/22
Address	Rävala pst 10, 10143 Tallinn (Estonia)
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E-mail	ioannis.gialamas@kbfi.ee & gialamas93@gmail.com

Profile in iNSPIRE

Profile in Google Scholar

ORCID ID

Research Positions

2022–Present **Postdoctoral Researcher**, *Laboratory of High Energy and Computational Physics*, National Institute of Chemical Physics and Biophysics (NICPB), Tallinn.

2021–2022 **Research Associate**, *Physics Department*, National and Kapodistrian University of Athens (NKUA), Athens.

Short-term visiting positions

08/2025 **Visitor**, *CERN*, Department of Theoretical Physics , Geneva.

10/2023 **Visitor**, *Austrian Academy of Science*, Institute of High Energy Physics (HEPHY), Vienna.

Research Interests

- Dynamical dark energy
- Euclidean wormholes and wave function of the Universe
- Massive spin-2 theories
- Cosmic inflation in modified theories of gravity
- Electroweak vacuum stability and primordial black holes
- Supergravity and gravitino dark matter
- Scale-Invariant Quadratic Gravity and applications to cosmic inflation

Education

2017-2021 **PhD in Theoretical Particle Physics**, *National and Kapodistrian University of Athens*, Department of Physics, Athens.
Grade Excellent

2015-2017 **MSc in Astrophysics–Astronomy–Mechanics**, *National and Kapodistrian University of Athens*, Department of Physics, Athens.
Grade 9.78/10.00 (Excellent)

2011-2015 **BSc in Physics**, *National and Kapodistrian University of Athens*, Department of Physics, Athens.
Grade 8.52/10.00 (Excellent)

PhD Thesis

Title *Cosmological Implications of Theories Beyond the Standard Model of Particle Physics* *pdf*
Supervisor Associate Professor Vassilis Spanos

Master Thesis

Title *Higgs-field tunneling in the background of black holes* *pdf in greek*
Supervisor Professor Nikolaos Tetradis

Bachelor thesis

Title *General Theory of Relativity, Cosmology and Applications*
Supervisor Associate Professor Theodosios Christodoulakis

Publications

31 P. Betzios, P. Ghiringhelli, I. D. Gialamas, and O. Papadoulaki, "A Menagerie of Wormholes and Cosmologies in the Gravitational Path Integral," arXiv:2602.23432 [hep-th].

30 I. D. Gialamas, "Reheating in geometric Weyl-invariant Einstein-Cartan gravity," arXiv:2602.00317 [gr-qc].

Postdoc → 4th year

29 I. D. Gialamas, G. Hütsi, M. Raidal, J. Urrutia, M. Vasar, and H. Veermäe, "Quintessence and phantoms in light of DESI 2025," Phys.Rev.D **112** (2025) 6, 063551, arXiv:2506.21542 [astro-ph.CO].

28 I. D. Gialamas, T. Katsoulas and K. Tamvakis, "Towards a gravitational network: Bridging metric-affine gravity and no-scale supergravity," Phys.Rev.D **112** (2025) 9, 9, arXiv:2506.02159 [hep-th].

27 I. D. Gialamas, T. Katsoulas and K. Tamvakis, "Keeping the relation between the Starobinsky model and no-scale supergravity ACTive," JCAP **09** (2025) 060, arXiv:2505.03608 [gr-qc].

26 I. D. Gialamas, A. Karam, A. Racioppi, M. Raidal, "Has ACT measured radiative corrections to the tree-level Higgs-like inflation?," Phys.Rev.D **112** (2025) 10, 103544, arXiv:2504.06002 [astro-ph.CO].

25 Eleonora Di Valentino et al., "The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics," Phys.Dark Univ. 49 (2025) 101965, arXiv:2504.01669 [astro-ph.CO].

24 C. Balazs et al. [*Linear Collider Vision Collaboration*], "The Linear Collider Facility (LCF) at CERN," arXiv:2503.24049 [hep-ex].

23 D. Attié et al. [*Linear Collider Vision Collaboration*], "A Linear Collider Vision for the Future of Particle Physics," arXiv:2503.19983 [hep-ex].

22 I. D. Gialamas and K. Tamvakis, "Dynamically induced spin-2 mass in a Weyl-invariant framework," Phys.Rev.D **112** (2025) 8, 084051, arXiv:2503.16598 [hep-th].

21 I. D. Gialamas and A. Racioppi "Symmetry-breaking inflation in non-minimal metric-affine gravity," JCAP **05** (2025) 072, arXiv:2412.17738 [gr-qc].

20 P. Betzios, I. D. Gialamas, O. Papadoulaki "Magnetic Anti-de Sitter Wormholes as seeds for Higgs Inflation," Phys.Rev.D **111** (2025) 12, 123542, arXiv:2412.03639 [hep-th].

Postdoc → 3rd year

19 I. D. Gialamas and K. Tamvakis "Inflation in Weyl-invariant Einstein-Cartan gravity," Phys.Rev.D **111** (2025) 4, 044007, arXiv:2410.16364 [gr-qc].

18 H. Eberl, I. D. Gialamas and V. C. Spanos, "Gravitino Thermal Production, Dark Matter, and Reheating of the Universe," JCAP **01** (2025) 079, arXiv:2408.16043 [hep-ph].

17 I. D. Gialamas, G. Hütsi, K. Kannike, A. Racioppi, M. Raidal, M. Vasar, H. Veermäe, "Interpreting DESI 2024 BAO: late-time dynamical dark energy or a local effect?," Phys.Rev.D **111** (2025) 4, 043540 arXiv:2406.07533 [astro-ph.CO].

16 I. D. Gialamas, Timo J. Karkkainen and Luca Marzola, "Generalized uncertainty principle and neutrino phenomenology," Phys.Lett.B 856 (2024) 138880, arXiv:2405.20925[hep-ph].

15 I. D. Gialamas, T. Katsoulas and K. Tamvakis, "Inflation and reheating in quadratic metric-affine gravity with derivative couplings," JCAP **06** (2024) 005, arXiv:2403.08530 [gr-qc].

14 I. D. Gialamas and K. Tamvakis, "On the absence of ghosts in quadratic bigravity," JCAP **03** (2024) 016, arXiv:2311.14799[gr-qc].

Postdoc → 2nd year

13 I. D. Gialamas and K. Tamvakis, "Bimetric Starobinsky Model," Phys.Rev.D **108** (2023) 10, 104023, arXiv:2307.05673[gr-qc].

12 I. D. Gialamas, H. Veermäe, "Electroweak vacuum decay in metric-affine gravity," Phys.Lett.B 844 (2023) 138109, arXiv:2305.07693 [hep-th]

11 I. D. Gialamas, A. Karam, T. D. Pappas and E. Tomberg, "Implications of Palatini gravity for inflation and beyond," Int.J.Geom.Meth.Mod.Phys. (2023), arXiv:2303.14148 [gr-qc].

10 I. D. Gialamas and K. Tamvakis, "Bimetric-Affine Quadratic Gravity," Phys. Rev. D **107**, no.10, 104012 (2023) arXiv:2303.11353 [gr-qc].

9 I. D. Gialamas and K. Tamvakis, "Inflation in Metric-Affine Quadratic Gravity,"JCAP **03**, 042 (2023), arXiv:2212.09896 [gr-qc].

8 I. D. Gialamas, A. Karam and T. D. Pappas, "Gravitational corrections to electroweak vacuum decay: metric vs. Palatini," Phys.Lett.B 840 (2023) 137885, arXiv:2212.03052 [hep-ph].

Postdoc → 1st year

7 I. D. Gialamas, A. Karam, T. D. Pappas and V. C. Spanos, "Scale-Invariant Quadratic Gravity and Inflation in the Palatini Formalism," Phys. Rev. D **104**, no.2, 023521 (2021), arXiv:2104.04550 [astro-ph.CO].

6 H. Eberl, I. D. Gialamas and V. C. Spanos, "Gravitino thermal production revisited," Phys. Rev. D **103**, no.7, 075025 (2021), arXiv:2010.14621 [hep-ph].

5 I. D. Gialamas, A. Karam, A. Lykkas and T. D. Pappas, "Palatini-Higgs inflation with non-minimal derivative coupling," Phys. Rev. D **102**, no.6, 063522 (2020), arXiv:2008.06371 [gr-qc].

4 I. D. Gialamas, A. Karam and A. Racioppi, "Dynamically induced Planck scale and inflation in the Palatini formulation,"JCAP **11**, 014 (2020), arXiv:2006.09124 [gr-qc].

3 I. D. Gialamas and A. Lahanas, "Reheating in R^2 Palatini inflationary models," Phys. Rev. D **101**, no.8, 084007 (2020), arXiv:1911.11513 [gr-qc] .

2 D. D. Canko, I. D. Gialamas and G. P. Kodaxis, "A simple $F(R, \phi)$ deformation of Starobinsky inflationary model," Eur. Phys. J. C **80**, 458 (2020), arXiv:1901.06296 [hep-th].

PhD →

1 D. Canko, I. Gialamas, G. Jelic-Cizmek, A. Riotto and N. Tetradis, "On the Catalysis of the Electroweak Vacuum Decay by Black Holes at High Temperature," Eur. Phys. J. C **78**, 328 (2018), arXiv:1706.01364 [hep-th].

Master →

Conference Proceedings

3 I. D. Gialamas and K. Tamvakis, "Cosmic inflation in metric-affine gravity," HEP 2023, Ioannina, Greece, arXiv:2107.09319 [hep-ph]

2 I. D. Gialamas, A. Karam, T. D. Pappas. A. Racioppi and V. C. Spanos, "Scale-invariance, dynamically induced Planck scale and inflation in the Palatini formulation," HEP 2021, Thessaloniki, Greece, J. Phys. Conf. Ser. **2105**, no.1, 012005 (2021), arXiv:2410.22263 [gr-qc]

1 H. Eberl, I. D. Gialamas and V. C. Spanos, "Gravitino thermal production," BSM 2021, (CFP) Zewail City of Science and Technology, Egypt, arXiv:2107.09319 [hep-ph]

Academic Activity

01/04/2018–08/04/2018 **Judge in the 10th International Physicists' tournament and team leader of the greek team** , *Moscow Institute of Physics and Technology (MIPT)*, Moscow.

2017–2018 **Mentoring the team of the University of Athens for participation in the International Physicists' tournament**, *Moscow Institute of Physics and Technology (MIPT)*, Moscow.

2016–2017 **Teaching assistant in the undergraduate course High Energy Astrophysics**, *National and Kapodistrian University of Athens (NKUA)*, Athens.

Research Groups and Societies

2025–Present **LISA Consortium**, [webpage].

2024–Present **Center of Excellence Fundamental Universe (CoE Universum)**, Estonia [webpage].

2022–Present **The European Consortium for Astroparticle Theory (EuCAPT)**, [webpage].

2021–Present **Hellenic Society for the Study of High Energy Physics**, *National Technical University of Athens (NTUA)*, Athens [webpage].

2017–2021 **Theoretical Particle Physics beyond the Standard Model Group**, *National and Kapodistrian University of Athens (NKUA)*, Athens [webpage].

Referee activity for international journals

Journal of Cosmology and Astroparticle Physics.

Physical Review Letters.

Physical Review D.

Physics Letters B.

International Journal of Modern Physics.

General Relativity and Gravitation.

Talks (selected)

24/09/2025 **Anti-de Sitter Wormholes as seeds for Higgs Inflation**, *Scalars 2025*, Warsaw, Poland.

03/07/2025 **Massive spin-2 particles: Bimetric gravity and its applications to cosmology**, *HEP 2025 - 41st Conference on Recent Developments in High Energy Physics and Cosmology*, Athens, Greece.

10/09/2024 **Massive Spin-2 Particles: An Introduction to Bimetric Gravity and Cosmology**, *12th Aegean Summer School*, Sifnos, Greece.

22/12/2023 **Corrections to electroweak vacuum decay in metric-affine gravity**, *Xmas Theoretical Physics Workshop 2023*, Athens, Greece.

20/06/2023 **Corrections to electroweak vacuum decay in metric-affine gravity**, *Geometric Foundations of Gravity 2023*, Tartu, Estonia.

15/05/2023 **Bimetric gravity and its extension to metric-affine gravity**, *Invited talk at Institute of Physics, Silesian University*, Opava, Czech Republic.

06/12/2022 **Gravitational effects to electroweak vacuum decay**, *Invited talk at Institute of Physics, Silesian University*, Opava, Czech Republic.

05/12/2022 **Quadratic gravity: Inflation and Reheating in the Palatini formulation**, *Invited talk at Institute of Physics, Silesian University*, Opava, Czech Republic.

29/06/2022 **Dynamically induced Planck scale and inflation in Palatini quadratic gravity**, *SUSY 2022 - The XXIX International Conference on Supersymmetry and Unification of Fundamental Interactions*, Ioannina, Greece.

28/06/2021 **Thermal production of gravitinos**, *Doctoral Candidates Conference at the National and Kapodistrian University of Athens*, Athens, Greece.

17/06/2021 **Scale-invariance, dynamically induced Planck scale and inflation in the Palatini formulation**, *HEP 2021 - 38th Conference on Recent Developments in High Energy Physics and Cosmology 16-19 June 2021*, Thessaloniki, Greece.

11/05/2021 **Quadratic gravity and inflation in the Palatini formalism**, *Seminar in the Journal Club of National and Kapodistrian University of Athens* , Athens, Greece.

13/06/2019 **Thermal production of gravitinos**, *Doctoral Candidates Conference at the National and Kapodistrian University of Athens*, Athens, Greece.

06/06/2018 **Thermal production of gravitinos**, *Doctoral Candidates Conference at the National and Kapodistrian University of Athens*, Athens, Greece.

Conferences and Workshops (selected)

22/09/2025–25/09/2025 **Scalars 2025**, *Warsaw*, Poland.

01/07/2025–04/07/2025 **HEP 2025-41st Conference on Recent Developments in High Energy Physics and Cosmology**, *Athens*, Greece.

02/06/2025–06/06/2025 **12th LISA Cosmology Working Group Workshop**, *Tallinn*, Estonia.

09/09/2024–14/09/2024 **12th Aegean Summer School**, *Sifnos*, Greece.

21/12/2023–22/12/2023 **Xmas Theoretical Physics Workshop**, *Athens*, Greece.

19/06/2023–22/06/2023 **Geometric Foundations of Gravity 2023**, *Tartu*, Estonia.

27/06/2022–02/07/2022 **SUSY 2022**, *Ioannina*, Greece.

16/06/2021–19/06/2021 **HEP 2021-38th Conference on Recent Developments in High Energy Physics and Cosmology**, *Thessaloniki*, Greece.

29/03/2021–02/04/2021 **Beyond Standard Model: From Theory to Experiment (BSM- 2021)**, *Egypt.*

19/12/2019–20/12/2019 **Xmas Theoretical Physics Workshop**, *Athens*, Greece.

31/08/2019–11/09/2019 **Workshop on Conecting Insights in Fundamental Physics: Standard Model and Beyond**, *Corfu*, Greece.

17/04/2019–20/04/2019 **HEP 2019 Conference on Recent Developments in High Energy Physics and Cosmology**, *Athens*, Greece.

20/12/2018–21/12/2018 **Xmas Theoretical Physics Workshop**, *Athens*, Greece.

20/12/2017–21/12/2017 **Xmas Theoretical Physics Workshop**, *Athens*, Greece.

International Schools

31/08/2019–11/09/2019 **CORFU2019: 19TH Hellenic School and Workshops on Elementary Particle Physics and Gravity**, *Corfu*, Greece.

03/06/2019–10/06/2019 **BS2019: SEENET-MTP Balkan School on High Energy and Particle Physics: Theory and Phenomenology**, *Ioannina*, Greece.

Scholarships and Awards

01/02/2019 **Award and Grant from Nuclear and Particle Physics section of University of Athens for PhD students**.

24/04/2018–24/04/2021 **Scholarship from the State Scholarships Foundation (I.K.Y.) for PhD students** .

01/09/2017–31/08/2018 **Scholarship from A.G. Leventis Foundation for PhD students**.

2018 **Fellowship from the Hellenic Foundation for Research and Innovation (HFRI)**, *2nd HFRI Call for PhD Candidate Fellowships*, (not utilized).

Personal grants (after PhD)

01/01/2026– **Starting Grant from the Estonian Research Council**, *Title: The Chronicles of Axions: From Inflation to Production, Project Number: PSG1132*, (Tallinn), PI: Ioannis Gilamas.

01/12/2023–30/11/2025 **Incoming Postdoctoral Grant by the Estonian Research Council**, *Title: Solving Challenges in Cosmology and Particle Physics, Supervisor: Martti Raidal, Project Number: MOBJD1202*, (Tallinn), PI: Ioannis Gialamas.

01/10/2022–30/09/2024 **Incoming Postdoctoral Grant by the Estonian Research Council**, *Title: Testing Cosmology and Particle Physics with Gravitational Waves, Supervisor: Martti Raidal, Project Number: SJD18*, (Tallinn), PI: Ioannis Gialamas.

Conference Participation Grants

21/09/2025–26/09/2025 **Conference grant from COST Action CA21136**, *Conference: Scalars 2025: Higgs Bosons and Cosmology*, Warsaw.

31/08/2019–11/09/2019 **Conference grant from COST Action CA15108**, *Conference: Workshop on Connecting Insights in Fundamental Physics: Standard Model and Beyond*, Corfu.

Team grants

01/01/2024–31/12/2030 **Center of Excellence grant**, *Title: Fundamental Universe - Universum, Project Number: TK202*, (Tallinn), PI: Martti Raidal.

01/01/2023–31/12/2025 **CERN**, *Title: CERN Science Consortium of Estonia, Project Number: RVTT3*, (Tallinn), PI: Martti Raidal.

01/01/2023–31/12/2025 **ESA**, *Title: ESA Science Consortium of Estonia, Project Number: RVTT7*, (Tallinn), PI: Hardi Veermäe.

01/04/2020–31/03/2023 **First Call for H.F.R.I Research Projects to Support Faculty Members & Researchers and Procure High-Value Research Equipment**, *Title: Challenges in Cosmology and Astroparticle Physics, Project Number: 824*, (Athens), PI: Nikolaos Tetradis.

Languages

Greek	Native language
English	Fluent
Italian	Beginner

Computer skills

- Microsoft Office
- Fortran
- Mathematica
- LaTeX
- Matlab
- C
- Python
- Javascript, HTML (in hobby projects)

Other Interests

- Playing football (as a goalkeeper) in many amateur teams in Athens, e.g. AC Patisia
- I enjoy running in long distance marathon
- Playing the greek instrument bouzouki

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